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I. Introduction

For the second semester, the Ponderosa High School (PHS) sustainability team will be finalizing the project and improving the renewable energy system at the PHS greenhouse. This document summarizes the work done in the spring semester and the work to be completed this semester. The human powered bicycle generation is analyzed separately from the greenhouse system. The greenhouse system looks at wind and solar energy, battery storage, tortoise enclosure, grow light pulleys, and switches/outlets. The bike system analyzes energy generation, electrical work, and coding.

II. Top Level Design Summary

The goal of the PHS sustainability project is to analyze and improve the off-grid renewable energy system located at the Ponderosa High School (PHS) greenhouse.

a. The Greenhouse

The greenhouse segment focuses on improving the overall power generation and storage in the off-grid renewable energy system. This is accomplished through items such as improved solar generation, increased battery storage capacity, and thermal regulation of the greenhouse. The greenhouse segment also includes smaller tasks such as new switches and outlets, a grow light pulley system, and a tortoise house enclosure.

Presented below are the customer and engineering requirements for the greenhouse segment of the PHS sustainability project. There is no CAD provided for the greenhouse because the items being added to the greenhouse are described below in the flow charts and additional diagrams. The items for the greenhouse are pre-manufactured and installed to the greenhouse, compared to the human powered bicycle which is manufactured by the team.

Table I

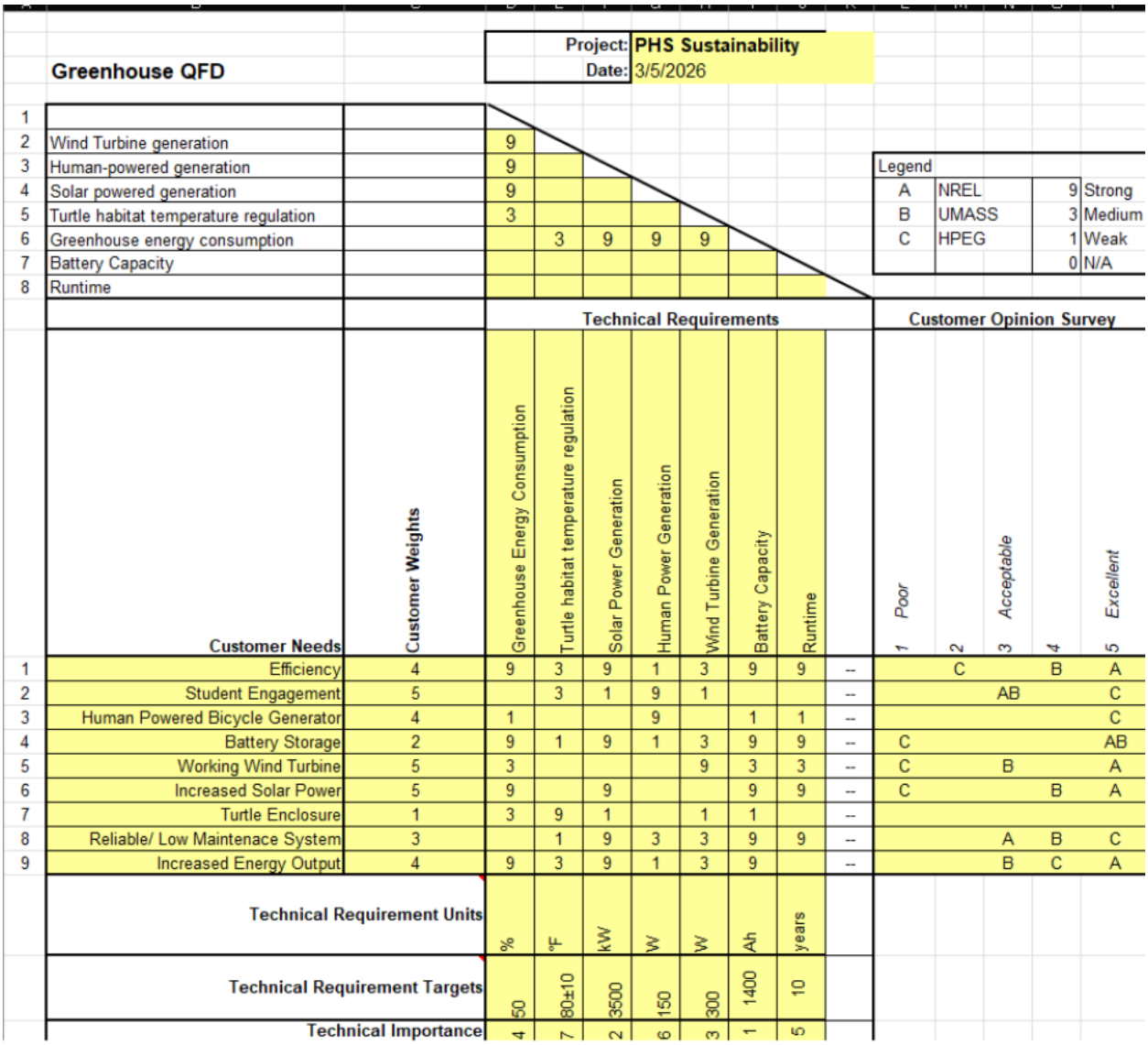
Customer Requirements

CR#	Customer Requirement	Description
1	Efficiency	The renewable energy system should operate efficiently by maximizing the conversion of available solar, wind, and human-generated energy into electrical power while minimizing energy losses in generation, storage, and distribution.
2	Student Engagement	The system should be interactive and educational so that PHS students can learn about renewable energy concepts. The project should include visual demonstrations, real-time monitoring, and hands-on learning opportunities that support STEAM education.
3	Human Powered Generation	The system should include a human powered generation station where PHS students can interact and produce renewable energy. The human powered generation system must fit inside the greenhouse and interact with the renewable energy system.
4	Battery Storage	The system must include sufficient batter storage capacity to store renewable energies, solar, wind, and human power, along with supplying power to the greenhouse loads when energy generation is unavailable. The battery system should support reliable operation and maintain safe limits.
5	Working Wind Turbine	The existing wind turbine has been inspected and verified that it is operating correctly. The turbine will generate measurable electrical power under appropriate wind conditions and will contribute to the overall renewable energy supply.
6	Increased Solar Power	The system will increase solar energy generation by adding additional panels to meet the demands of the greenhouse.
7	Tortoise Enclosure	The greenhouse contains a tortoise that is unable to live in the greenhouse year-round due to the low temperatures in the winter. The system will use lighting, heating, a new enclosure, and water circulation to regulate this temperature. The energy system must reliably support these loads to maintain proper living conditions.
8	Reliable/low Maintenance System	The renewable energy system should be designed to operate reliably with minimal maintenance requirements. Components should be durable, easy to operate, and suitable for a school environment where technical maintenance resources may be limited.
9	Increased Energy Output	The system should generate enough energy to support the current greenhouse loads while allowing for additional features and expansion of educational equipment.

Table II
Engineering Requirements

ER#	Engineering Requirement	Description	Target	Units
1	Greenhouse Energy Consumption	Percentage of greenhouse electrical demand supplied by the renewable energy system	50	%
2	Tortoise Habitat Temperature Regulation	Maintain a stable temperature range suitable for the tortoise habitat environment	80±5	Fahrenheit
3	Solar powered generation	Total electrical power generated by the system	3500	kW
4	Human Power Generation	Electrical power produced by the bicycle(s) during operation	100	W
5	Wind Turbine Generation	Electrical power output generated by the wind turbine	300	W
6	Battery Capacity	The maximum storage of energy in the batteries	400	Ah
7	Runtime	The time the batteries can run until they die or are less than 60% efficient	10	years

The greenhouse QFD provides a technical overview of customer and engineering requirements, showing the relationship between all CRs and ERs. Target units for engineering requirements are provided below in the QFD.



Greenhouse QFD

Project: **PHS Sustainability**
Date: 3/5/2026

1																
2	Wind Turbine generation		9													
3	Human-powered generation		9													
4	Solar powered generation		9													
5	Turtle habitat temperature regulation		3													
6	Greenhouse energy consumption			3	9	9	9									
7	Battery Capacity															
8	Runtime															

Legend			
A	NREL	9	Strong
B	UMASS	3	Medium
C	HPEG	1	Weak
		0	N/A

			Technical Requirements								Customer Opinion Survey				
	Customer Needs	Customer Weights	Greenhouse Energy Consumption	Turtle habitat temperature regulation	Solar Power Generation	Human Power Generation	Wind Turbine Generation	Battery Capacity	Runtime		1 Poor	2	3 Acceptable	4	5 Excellent
1	Efficiency	4	9	3	9	1	3	9	9	--		C		B	A
2	Student Engagement	5		3	1	9	1			--			AB		C
3	Human Powered Bicycle Generator	4	1			9		1	1	--					C
4	Battery Storage	2	9	1	9	1	3	9	9	--	C				AB
5	Working Wind Turbine	5	3				9	3	3	--	C		B		A
6	Increased Solar Power	5	9		9			9	9	--	C			B	A
7	Turtle Enclosure	1	3	9	1		1	1		--					
8	Reliable/ Low Maintenace System	3		1	9	3	3	9	9	--			A	B	C
9	Increased Energy Output	4	9	3	9	1	3	9		--			B	C	A
	Technical Requirement Units		%	°F	kW	W	W	Ah	years						
	Technical Requirement Targets		50	80±10	3500	150	300	1400	10						
	Technical Importance		4	7	2	6	3	1	5						

Fig 1. Greenhouse QFD

b. The Human Powered Bicycle

The human powered bicycle segment focuses on the manufacture and installation of a human powered bicycle. The goal of the human powered bicycle is to engage PHS students in STEAM education and the off-grid renewable energy system with the interactive design. Additionally, the human-powered bicycle will provide additional power to the greenhouse for needed demands.

Below is the updated CAD model for the human powered bicycle. A new update from the spring semester of the PHS sustainability capstone is that the stationary bike frame will be purchased and installed instead of manufactured. This is due to a $1500 increase in the budget.

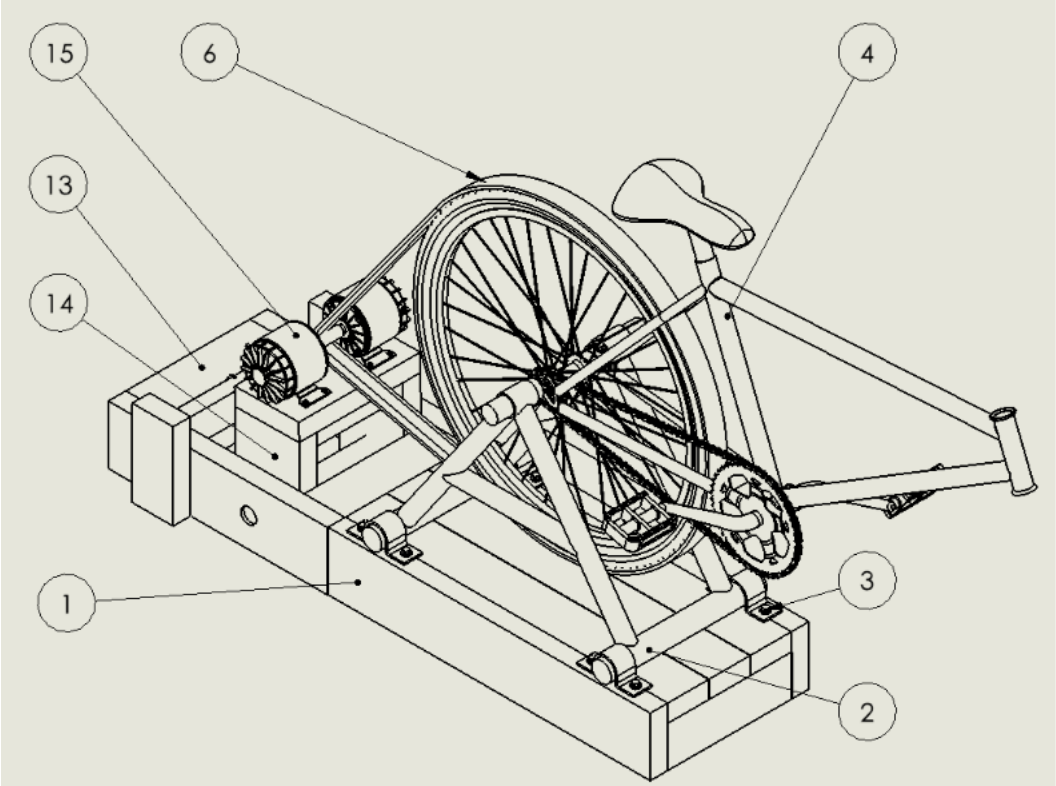


Fig 2. Drawing of CAD Model with listed components

The overall design of the Frame will stay the same with the bicycle being placed upon a manufactured base to ensure that the bicycle stays level to the ground, since the ground of the greenhouse is gravel and prone to shifting with movement. Components 1 and 13 are the wooden bases that bicycle frame, alternator and overall system will sit on. Component 2 is the bicycle frame, which the team originally planned to weld together using the NAU Machine shop, but thanks to a previously mentioned increase in budget the team plans on purchasing a premade one. Component 4 is the bicycle that the students will be pedaling to generate power, with Component 2 holding the bicycle stand in place. Component 6 is the belt that is connected to component 15, alternator, which will generator energy to the greenhouse electrical system. Component 14 is a tension spring system to ensure that the belt stays in tension with the bicycle wheel and alternator to minimize potential energy loss.

Provided below are the customer and engineering requirements for the human powered bicycle segment of the PHS sustainability project.

Table III

Customer Requirements

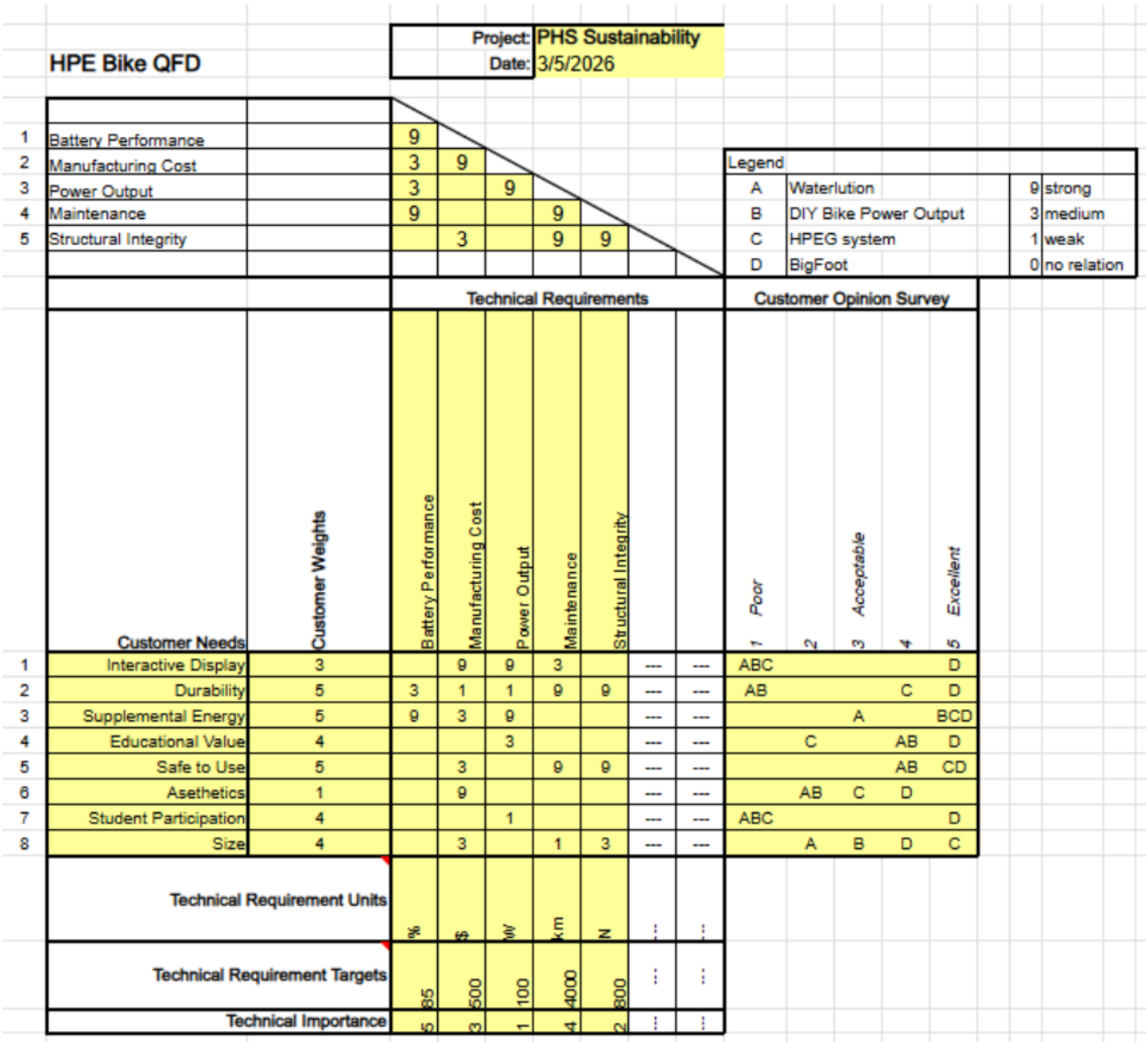
CR#	Customer Requirement	Description
1	Interactive Display	An interactive display will showcase the energy generated, the total energy generation from all time, and voltage output. The goal of the interactive display is to inform students on power and energy generation.
2	Durability	This bike is designed to be left alone for years in an outdoor setting with minimal maintenance needed. It should be able to withstand the

		dry heat, additional moisture in the greenhouse, and student use for five to ten years.
3	Supplemental Energy	This bike is not intended to be a main power source, but a small supplement of energy for cloudy days or during the evening. The students should be able to charge the batteries for small loads around the greenhouse.
4	Educational Value	This bike is used as a visual and physical representation of energy. The students will be able to cycle and see their energy generation through an interactive display. This information will be explained through multiple student instructions provided by the PHS team.
5	Safe to Use	The bike supports/truss will be made from steel to ensure stability and reduce the risk of fracture. The alternator will be protected by thick plastic box so the students cannot accidentally injure themselves.
6	Aesthetics	The bike will look safe to use and have a human centric design, so the bike is used for its intended purpose.
7	Student Participation	The aspect of a reward, like lighting up the fish tank, will encourage students to participate and use the bike. Making the bike easy to ride, having minimal resistance, and a purposeful output are necessary for student participation.
8	Size	The bike will be stored in the greenhouse and must fit in the allowed space from the clients.

Table IV
Engineering Requirements

ER#	Engineering Requirement	Description	Target	Units
1	Battery Performance	The efficiency and power output delivered from the battery	85	%
2	Manufacturing Cost	The cost to buy and assemble the bike stand	500	$
3	Power Output	Power generated by the bike and converted into usable energy	100	W
4	Maintenance	The distance the motor can run until it breaks down	4000	Km
5	Structural Integrity	The weight that the bike stand can hold	800	N

The QFD for the human powered bicycle segment of the PHS sustainability project is provided. The QFD gives a technical overview of the customer requirements, engineering requirements, and the relationship between them. Target units for the human powered bicycle are listed below.



HPE Bike QFD

Project: PHS Sustainability
Date: 3/5/2026

1	Battery Performance		9						
2	Manufacturing Cost		3	9					
3	Power Output		3		9				
4	Maintenance		9			9			
5	Structural Integrity			3		9	9		

Legend			
A	Waterlution	9	strong
B	DIY Bike Power Output	3	medium
C	HPEG system	1	weak
D	BigFoot	0	no relation

		Technical Requirements							Customer Opinion Survey				
	Customer Needs	Customer Weights	Battery Performance	Manufacturing Cost	Power Output	Maintenance	Structural Integrity	---	---	1 Poor	2	3 Acceptable	4	5 Excellent
1	Interactive Display	3		9	9	3		---	---	ABC				D
2	Durability	5	3	1	1	9	9	---	---	AB			C	D
3	Supplemental Energy	5	9	3	9			---	---			A		BCD
4	Educational Value	4			3			---	---		C		AB	D
5	Safe to Use	5		3		9	9	---	---				AB	CD
6	Asethetics	1		9				---	---		AB	C	D	
7	Student Participation	4			1			---	---	ABC				D
8	Size	4		3		1	3	---	---		A	B	D	C
	Technical Requirement Units		%	$	W	km	N	...	...					
	Technical Requirement Targets		85	500	100	4000	800	...	...					
	Technical Importance		5	3	1	4	2	...	...					

Fig 3. The human powered bicycle QFD

III. Summary of Standards, Codes, and Regulations

Below is a summary of all standards, codes, and regulations that relate to the manufacturing, installation, and testing of sub systems in the PHS sustainability capstone project.

a. Electrical Installation

National Electric Code, Article 690

The electrical modifications to the PHS greenhouse were evaluated with consideration of the National Electrical Code (NEC), with particular emphasis on Article 690 – Solar Photovoltaic Systems. Article 690 establishes requirements for photovoltaic system conductors, overcurrent protection, grounding, equipment, and interconnection of photovoltaic system components. These requirements are applicable because the greenhouse operates as an off-grid renewable energy system incorporating photovoltaic generation, battery storage, charge control, an inverter, and AC electrical loads.

Electrical installation considerations are important for the proposed addition of four receptacles and one light switch within the greenhouse. The electrical distribution system must account for overcurrent protection, grounding, equipment ratings, and the environmental conditions present in the greenhouse including humidity and temperature. The electrical work will be planned and documented by the team while final installation must comply with applicable NEC requirements and be completed by a qualified person. This ensures that electrical failures are reduced and safety precautions are followed.

Taking into consideration the additional solar panels being integrated into the greenhouse system, multiple codes will be needed to ensure the building is not only able to handle the additional weight but also avoid damaging the electrical system to maintain the students' safe learning space. For electrical codes, the group will need to follow the National Fire Protection Association (NFPA) codes, specifically the NFPA 70 or the National Electrical Code. Since electricity will be involved and the area/building will have occupants inside/nearby the NFPA 1 code or the Fire Protection code will also need to be followed to mitigate potential fire hazards as well as update any missing code requirements. These codes provide detailed requirements needed to ensure "property protection by providing a comprehensive, integrated approach to fire code regulation and hazard management." [1]

As the solar panels will be on the roof the team will need to check the roofs structural integrity to ensure that it can support the increased weight as potential rain and snow weights following ANSI and OSHA regulations. The ANSI regulations on roofs provide a detailed outline of equations for the structural analysis of a building with consideration of multiple facets such as potential environmental loads, live loads (people), and dead loads (roof weight, solar panel weight).

b. Bicycle Generator and Electrical Systems

The human-powered bicycle generator is a permanently installed electrical generating source wired to a stationary storage battery and located inside an occupied school building. Additionally, it is operated by students rather than by trained personnel. As such, three separate bodies of requirements govern it: the installation codes that apply to any permanently installed DC source and its battery, the product safety standards that apply to the equipment itself, and the workmanship and labeling standards that apply to how the team builds and installs it. Table V lists each requirement, the portion that applies, and the specific design decision it drives.

Because the greenhouse is off-grid and never parallels the utility, the interconnection standards that normally dominate a renewable energy project do not apply; the equivalent requirements come from NEC Article 710. Table VI records what was evaluated and scoped out, so those decisions are documented rather than silent.

Table V

Standards and Codes Applicable to the Bicycle Generator

Code or Standard	Applicable Components	Design Requirements
NFPA 70 (NEC), 2017 ed. [7] [8]	Art. 710 stand-alone systems; 445 generators; 240 overcurrent; 480 storage batteries; 250 grounding and bonding; 720 circuits under 50 V	Coconino County enforces the 2017 NEC and is adopting the 2023 edition, the design is checked against both [7][8]: - Article 445 treats the alternator as a generator, requiring a means of disconnecting and conductors rated at 115 % of nameplate current.

		- Article 240 requires overcurrent protection at the source of each conductor, with a DC interrupting rating above the bank's available short-circuit current. - Article 250 permits exactly one bond between the DC negative and the equipment grounding conductor, made at the battery. - Article 480 sets the disconnect, guarding, working space and ventilation rules for the AGM bank. - Article 720 sets a 12 AWG minimum for the sub-50 V branch.
Equipment listings [9][10][11] UL 1741 · UL 1973 · UL/IEC 62368-1	Charge controller, any replacement battery, and the display electronics	The charge controller must be UL 1741 listed, which rules out the unlisted charger module used during breadboarding. Any replacement battery must be listed for UL 1973, and display electronics follow UL/IEC 62368-1. These are purchasing requirements, not something the team can verify by test.
47 CFR Part 15 [12]	Subparts B and C: unintentional and intentional radiators	The ESP32 contains a Wi-Fi and Bluetooth radio, so the team must use a pre-certified module and leave its antenna and shielding unmodified.
ISO 20957-1 and [13], ISO 20957-10 [14]	Stationary training equipment, general safety, and exercise bicycles with a fixed wheel	The finished unit is stationary training equipment in a supervised setting. Requires stability under off-axis loading, no shear or entrapment points within the user's reach, and guarding of all moving parts. Specifically, the point where the roller meets the tire, since the wheel is continuously loaded by the alternator. This is what turns the alternator enclosure from an aesthetic choice into a dimensioned guard.
ANSI Z535.4 [15]	Product safety signs and labels	Defines the signal word, color and panel layout of the two required labels: WARNING for rotating parts and pinch points at the alternator guard, and CAUTION at S1against opening the disconnect under load.
29 CFR 1910 Subpart S, 29 CFR 1910.147, NFPA 70E [16][17][18]	Installation practice, lockout sequence, and DC electrical safety	Apply to the team and to school staff during build-out and maintenance: conductor protection through structure, approved boxes and fittings, and maintained working clearance at the bank. A written lockout sequence is required: open S1, open the battery disconnect, verify zero voltage at the controller terminals, which is what the disconnect placement is designed to permit. Incident energy at 24 V is low, but the bank's short-circuit current is not, so insulated tools and no conductive jewelry at the terminals.

Table VI

Standards Evaluated and Excluded

Code or Standard	Exclusion Reasoning
NEC Art. 706, NFPA 855 [73], [86]	Article 706 is scoped to energy storage above 60 V DC, and the bank is 24 V nominal, so Article 480 governs. NFPA 855 applies above the fire code's ESS thresholds, which the 5.376 kWh lead-acid bank falls under. Both return to scope if a lithium bank is installed.
ANSI/CAN/UL 2849 [19]	Covers electrical systems that propel an e-bike. This system generates rather than propels. Retained as a reference for bicycle-mounted wiring and strain relief, not as a compliance requirement.

Three requirements change the design rather than confirm it. NEC Articles 240 and 445 added the overcurrent device and disconnect between the alternator and the charge controller. NEC Article 250 established that the bicycle branch carries no ground of its own and references the existing bond at the bank. ISO 20957-10 converts the alternator enclosure into a guard with a defined nip point to cover.

IV. Summary of Equations and Solutions

In this section, a summary of all equations and solutions generated are provided below. These solutions are considering the “worst case scenarios” for both the greenhouse and human powered bicycle. It is important when designing for longer term use with little to no maintenance that the greenhouse and human powered bicycle can endure “worst case scenarios” and continue normal operation.

The greenhouse’s load case accounted for factors such as time of day, time of year, and weather. During the month of December, the greenhouse reports the lowest energy generation (snow, cloudy days, and winter sun) and has larger energy demands (heat lamps for the bunny and tortoise, grow lights for plants, extended light hours). The improved greenhouse system must be able to maintain power to all loads during the month of December therefore the load case is what the team used for analysis.

The human powered bicycle’s load case accounted for factors such as user weight, stress points, and electrical overloads. The bicycle must be able to hold the average adult since this bike will be used by teenagers and adults at PHS. The design must be able to withstand the weight at concentrated stress points from the additional weight and electrical must be wired with safety factors, so there is no overheating in the alternator, or interactive display.

a. Equations and solutions for load cases

Table VII

Equations and Solutions for Load Cases

Team Member	Category	Equation	Solution	Notes
Greenhouse				

Alex	Energy Output	$\frac{\lvert P' - P \rvert}{P} * 100$	20.94%	Increase in energy output from hybrid system
Carson	Solar Energy	$E = A * R * \eta$ $E = \eta_s * PSH$	Appendix A1-2	Examines the solar energy output from the current greenhouse and the improved design
Alex	Solar Energy	$+\uparrow \varepsilon F = 0$ $+\circlearrowleft M_a = 0$ $\sigma = \frac{M_{max} * c}{I}$	R=237.6 lbf 2.5847 kip*ft 10.34in	Confirms that the roof can structurally support 10 additional solar panels
Kaitlyn	Wind Energy	$A = \pi * R^2$ $P_{wind} = \frac{1}{2} * \rho A v^3$ $P_{turbine} = \frac{1}{2} * C_p \rho$ $\lambda = \frac{\omega R}{v}$ $f(v) = \frac{k}{c}(\frac{v}{c})^{k-}$ $E[P] = \int_0^\infty P(v) f(v)\, dv$	0.82m^2 Appendix A3-5 2.55W Appendix A6 136.05W	Used to calculate the energy and power that could be produced by the small-scale wind turbine.
Jenna	Tortoise Enclosure	$\dot{Q}_{net} - \dot{W}_{net} = \frac{dE_{cv}}{dt}$ $\dot{Q}_{loss} = \dot{Q}_{heater} = UA(T_{in} - T_g) + \dot{Q}_{evap} + \dot{Q}_{vent}$ $\dot{Q} = UA\Delta T$ $E = \frac{\dot{Q}(W) * t(hr)}{1000}$ $E_{bask} = lamp * time * 365$	Appendix AI-III	Used to calculate or the power and energy required to sustain the tortoise enclosure
Alex	Grow light Pulley System	$+\uparrow F_y = 0$ $+\circlearrowleft M_o = 0$ $\sigma = \frac{M_{max} * c}{I}$ $FS = \frac{\sigma'}{\sigma}$	T_1=61.8N M_{max}=146.88N 0.6436 Mpa 16	Used to confirm grow-light pulley system design for installation
Alex	Battery Comparison	$Q = \int i\, dt$ $Q_{kWh} = V * Q$ $i(A) = \frac{Q}{t}$	Appendix AIV	Used to compare different batteries
Jenna	Battery Heat Loss	$P = W_{human} * \eta_{sys}$ $I = \frac{P}{V}$ $\dot{Q}_{gen} = hA(T_s - T_\infty)$ $\dot{Q}_{gen} = I^2 R$	Appendix AV-VI	Calculate the heat loss from the battery and ensure safe temperatures.

		$Fo = \frac{kt}{\rho c L^2}$		
		$Bi = \frac{hL_c}{k}$		
		$L_c = \frac{V}{A_S}$		
		$\theta^* = \frac{T_i - T_\infty}{T_i - T_\infty}$		
Human Powered Bicycle Generator				
Ethan	Energy Output	$G_{bike} = \frac{N}{n}$	Appendix BI and B1	Force-cadence model on gear cassette used to select the operating gear and to size the wheel-to-roller contact drive.
		$G_{alt} = \frac{r}{r_{alt}}$		
		$\omega_{alt} = \omega G_{bike} G_{alt}$		
		$\tau = Fr$		
		$P_{mech} = \tau\omega$		
		$\eta_{sys} = \eta_{alt}\eta_{drive}\eta_{contact}\eta_{charge}$		
		$P = P_{mech}\eta_{sys}$		
Ethan	Net Power Output	$t_r = \Sigma t_w$	Appendix BII-III	Model of the complete electrical system, including the sensing and display's power draw. Used to determine the display refresh interval, and the resulting P_{net} of 139.9 W will be used in bicycle output testing.
		$I_{active} = (\Sigma i_a)(\frac{t_r}{T})$		
		$I_{standby} = (\Sigma i_s)(\frac{T - t_r}{T})$		
		$I_{avg} = I_{active} + I_{standby}$	1.213mA	
		$P_{avg} = I_{avg} * v_s$	4.005mW Arduino (247.5W)	
		$P_{net} = P_{out} - P_{avg}$	139.996W	
Carson	Bike Stress	$\tau_{crank} = F_{pedal} * r_{crank}$	25.5Nm	Used to confirm the structural safety of the human powered bicycle
		$\tau_{sprocket,R} = \tau_{sprocket,FR} * (1 - n_{loss})$	24.735	
		$F_{shaft,R} = \frac{\tau_{sprocket,R}}{r_{shaft,R}}$	2473.5N	
		$F_{stand} = \frac{F_{shaft,R}}{2}$	1236.75N	
		$W_{wheel} = \frac{g * (m_{rider} + m_{bike})}{2}$	402.21N	
		$W_{wheel,rear} = \frac{W_{wheel}}{2}$	201.105N	
		$F_1 = \frac{F_{shaft} + W_{wheel,R})}{2\cos(35)}$	1129.28N	
		$F_2 = \frac{F_1\cos(35)}{\cos(350}$	1129.28N	
		$F_s = F_1\sin(35) - F_2\sin(35)$	0N	
		$F_N = F_2\cos(25)$	925.05N	
		$F_b = F_2\sin(25)$	647.73N	

Kaitlyn	Alternator Analysis	$\eta = \frac{P_{out}}{P_{human}}$	Appendix B2-5	Evaluates required speed increase between bicycle and alternator, effects of gear ratio on generator voltage, and alternator performance.
		$RPM_{gen} = G * RPM_{pedal}$		
		$E = (k_e)(\omega)$		
		$L_{10} = (\frac{c}{p})^p$		
		$L_{10h} = \frac{L_{10} * 10^6}{60 * RPM_{gen}}$		Evaluates mechanical lifetime effects associated with increasing alternator speed.

b. Factor of Safety (FoS)

Table VIII

Minimum FoS Information

Sub System	Load Case Scenario	Material	Method of Calculating FoS	FoS
Grow-light pulley system	One 12.6kg grow-light system attached to the wooden support beam by rope pulley system	Wood, rope	Hand calculation Appendix C1	16
Solar panel increase	10 additional solar panels added to PHS greenhouse roof	Wood, solar panels	Hand calculation Appendix C2	1.16
HPE bike frame	One user 165lbs (70kg) pedaling on HPE Bike	Steel, wood	Hand calculation Appendix C3	1.82

V. Flows Charts and Additional Diagrams

The team created black box and decomposition models to assess the flow of energy in the greenhouse and help the team break down the project into components. For a broad view of the greenhouse, a black box model was created.

a. The Greenhouse

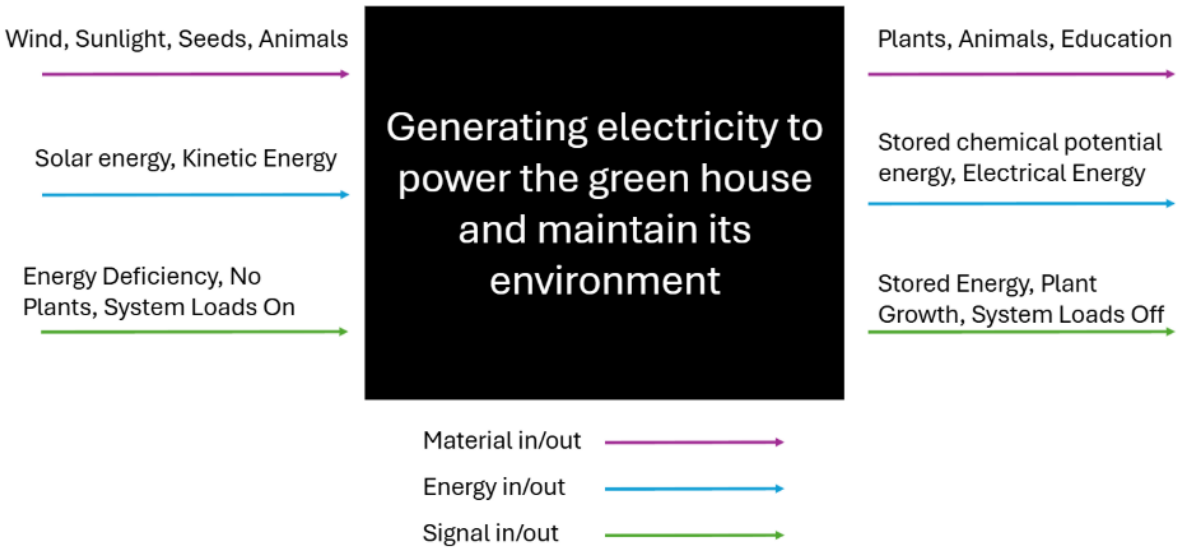


Fig 4. Greenhouse black box model

The model focuses on how the greenhouse maintains itself as an off-grid system. The material going into the system is sunlight, wind, and seeds. The material does change through the system as the sunlight and wind are converted into energy and the seeds sprout into plants, with the only material remaining the same being the animals as they serve as a method of maintaining the greenhouse's environment. Energy entering the system is a mix of solar and kinetic energy that is converted to stored chemical energy in the battery system and mechanical energy that goes and powers the greenhouse's loads. The signals that cause the system to function are the loads being active, battery storage being low on power and if no plants are growing. The systems turn off if the plants are growing, the battery storage is full, and the active loads are no longer active.

Having key components of the greenhouse modeled, each component can further be broken down into a decomposition model to see the flow path of energy through the system.

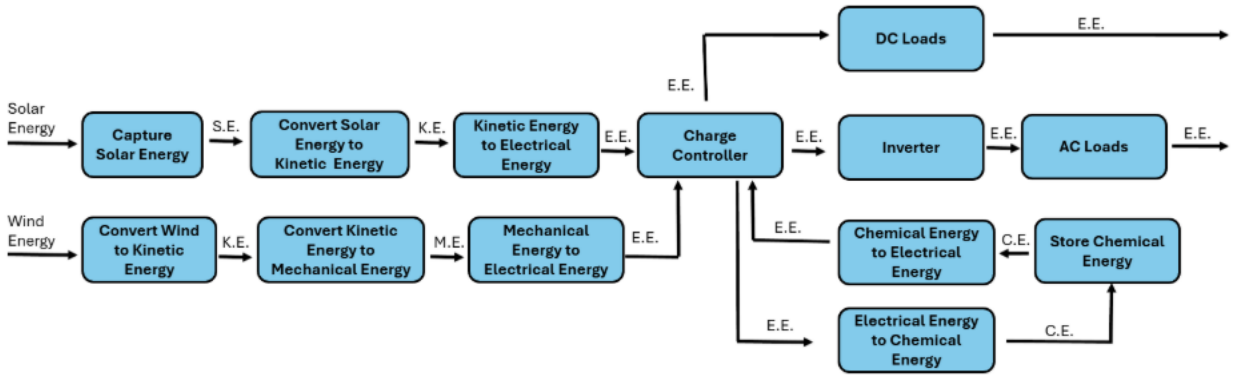


Fig 5. Decomposition model of the greenhouse

This model starts with both solar and wind energy, the two sources for the PHS greenhouse system. Their flows energies are converted from radiant and kinetic energy respectively, and then to electrical energy.

The electrical energy through a charge controller. The energy is distributed to battery storage by converting electrical energy to chemical energy, DC loads, and AC loads via an inverter. The chemical energy can go back into the charge controller by being converted back into electrical energy.

The decomposition of the greenhouse's energy system allows the team to connect the bicycle generator to the system and complete the energy audit. The team needs to understand how their system and each component maintains the greenhouse's off-grid system. This will serve as a guide for clearly defining each individual component of the grid.

b. The Human Powered Bicycle Generator

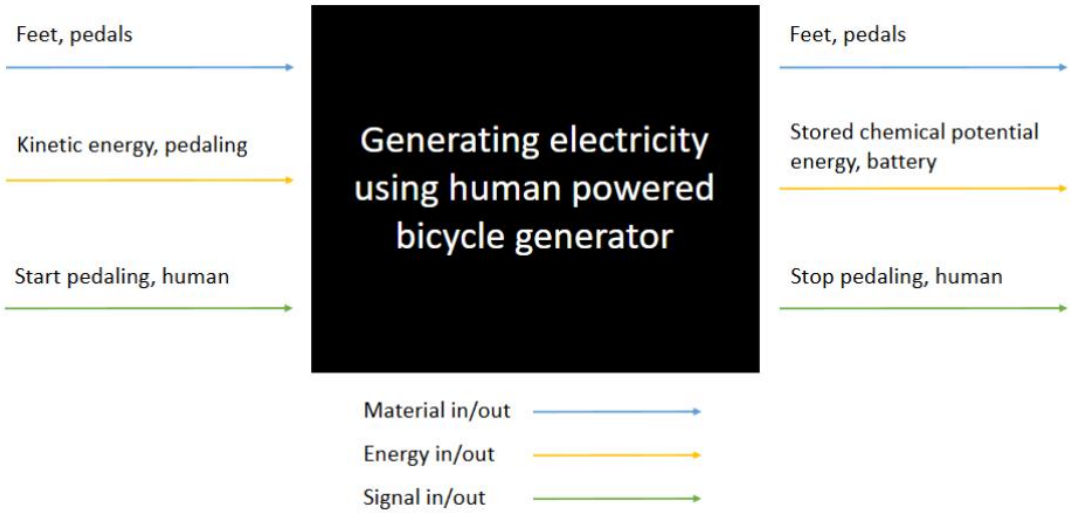


Fig 6. Black box model of the HPE Bike

Focusing on the bike, this model displays the electricity generation using the bicycle generator. The material going in/out of the bike is the feet rotating the pedals. The material does not change during the function. The energy going into the bike is the kinetic energy from pedaling which is then transformed into stored chemical potential energy in the PHS battery system. The signal going in/out of the bike is to start/stop pedaling.

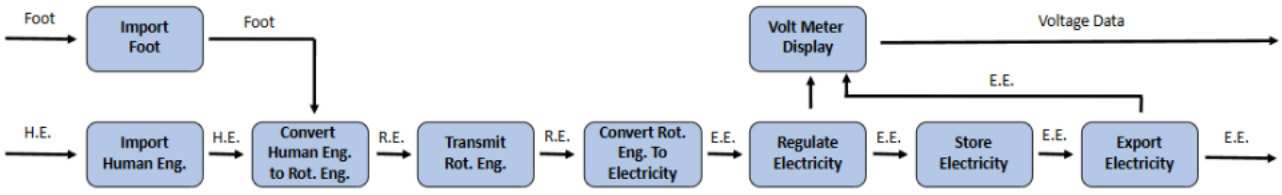


Fig 7. Decomposition model of the HPE bike

The decomposition model of the bike starts with pedaling to generate changes. This energy is converted to rotational energy which is then transmitted into electricity. To regulate and export electricity, a

voltmeter display is used to capture voltage data. The rest of the electricity generated is stored in the battery system.

The decomposition model showcases the necessary components for the team's concept generation. The components focused on are generating human energy (pedals, bike, etc.), a motor converting human energy into rotational energy, and a display of that energy (screen, light, etc.). The components identified in the decomposition model were used to generate concepts that later turned into a finalized design of the human powered bicycle generator.

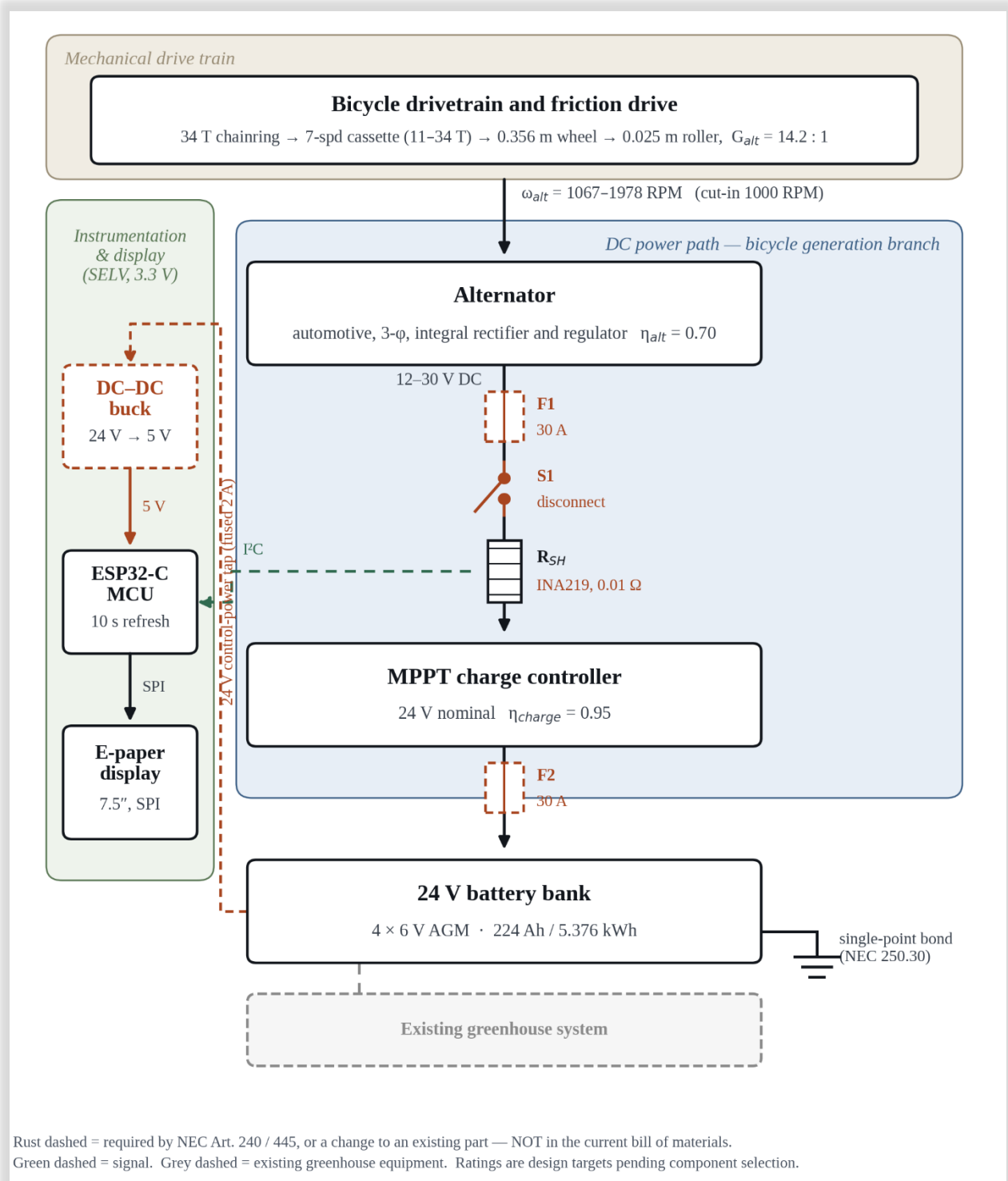


Fig 8. One-Line Power Diagram

Two diagrams describe the bicycle's electrical system. The one-line diagram above covers the whole branch, from the rider to the point where the bicycle ties into the existing battery bank, and carries the overcurrent devices, disconnect and grounding arrangement required by Section III. The circuit schematic below is a component-level pinout diagram of the sensing and display subsystem.

The mechanical group in the one-line carries the quantities that cross into the electrical domain: 125–250 N at the pedals at 45–75 RPM, a chainring-to-cassette ratio of 1.0 to 3.1, and a further 14.2:1 step-up from the wheel to the roller, giving an alternator shaft speed of 1067–1978 RPM against a 1000 RPM cut-in.

Table IX below details the system components in the one-line diagram.

Table IX

Power Line Components

Component	Function	Rating or value	Basis	Status
Alternator [G1]	Converts shaft rotation into DC. Rectification and voltage regulation are internal to the unit, so no external rectifier is required.	12–30 Vdc output η_{alt} = 0.70	ME 476C §5.4.6; existing system diagram	Donated
Fuse, alternator conductors [F1]	Protects the wiring between the alternator and the charge controller against a fault downstream of the alternator.	30 A ANL (placeholder)	Required by NEC 240.4 and 445.13. Final rating depends on the alternator nameplate, which has not been read.	Proposed
DC disconnect, 2-pole [S1]	Isolates the bicycle branch for service or lockout without shutting down the rest of the greenhouse system.	32 A (placeholder)	Required by NEC 445.18 (a generator needs a disconnecting means).	Proposed
Current shunt [U1 component]	Develops the small voltage the current monitor reads. Located in the positive conductor between S1 and the controller, so the reading is generated current rather than net battery current.	0.01 Ω, 3 W	The INA219 module ships with a 0.1 Ω shunt, which limits it to roughly ±3.2 A. Branch current is several times that, so the part must be swapped out.	Proposed
MPPT charge controller	Conditions the variable alternator output into a charging profile the 24 V bank accepts.	24 V nominal, η_{charge} = 0.95 (assumed)	Efficiency from ME 476C §5.4.6. UL 1741 listing is a purchasing requirement.	Unselected Component
Fuse, battery conductors	Protects the wiring between the controller and the	30 A ANL (placeholder)	Required by NEC 240.4. Final rating	Proposed

[F2]	battery bank against the bank's own short-circuit current, which is the larger hazard in a DC system.		depends on the bank's available short-circuit current, which has not been measured.	
24 V battery bank [BT1]	Stores generated energy. Shared with the solar and wind sources; the bicycle is a supplementary charging source.	4 × 6 V AGM in series; 224 Ah, 5.376 kWh	Existing system diagram	Current System
Tie-in to greenhouse system	Inverter/charger, solar and wind charge controllers, main disconnect and dump load. Out of scope for this branch; shown for context.	Victron MultiPlus-II 24/3000/70; 150 A main disconnect	Existing system diagram	Current System

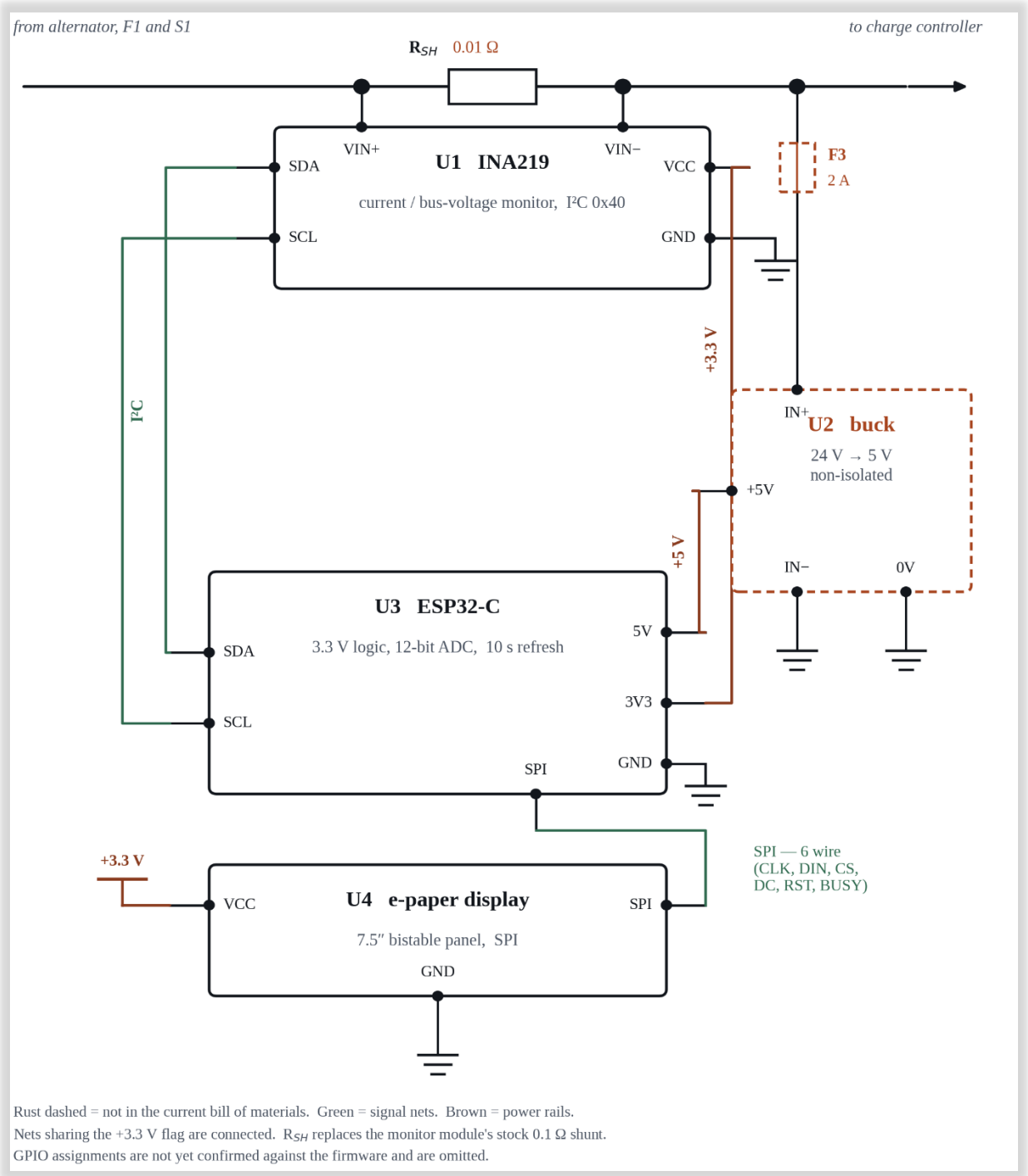


Fig 9. Sensing and Display Schematic

Table X below details the system components in the sensing and display circuit schematic.

Table X

Sensing and Display Components

Component	Function	Rating or value	Basis	Status
Control power fuse [F3]	Protects the small tap that feeds the display supply from the 24 V bus.	2 A (placeholder)	Required by NEC 240.4.	Proposed
DC–DC buck converter [U2]	Steps the 24 V bus down to 5 V for the logic.	24 V → 5 V, 1 A, non-isolated	Needed to power the microcontroller from the bus. The current prototype powers the board separately.	Proposed
INA219 current and voltage monitor [U1]	Measures charge current across the shunt and bus voltage, and reports both to the microcontroller.	I²C address 0x40, 12-bit; ±20 A with the 0.01 Ω shunt	Present in the prototype circuit	Current System
ESP32-C microcontroller [U3]	Reads the monitor, computes instantaneous power and cumulative energy, drives the display, and sleeps between refreshes.	3.3 V logic; I_{avg} = 1.213 mA at a 10 s refresh		Current System
E-paper display [U4]	Shows instantaneous power output and cumulative energy generated. Bistable, so it holds an image with no power between refreshes.	7.5″ panel; 0.001 mA standby	ME 476C Table XXV. The prototype currently uses a 7.5” module	Current System
Decoupling capacitor [C1]	Stabilizes the 5 V rail against the current step when the display refreshes.	100 nF	Standard practice.	Proposed
SPI interface	Six-wire link from the microcontroller to the display panel: clock, data, chip select, data/command, reset and busy.	CLK, DIN, CS, DC, RST, BUSY	Connection set from the prototype circuit.	Current System

Both new diagrams changed the design rather than recording it. The one-line added the two overcurrent devices and disconnected, and made the single-point bond explicit, which settled whether the stand needed a ground of its own. The schematic caught the shunt sizing error and fixed the point of measurement, which determines whether the display reads generated power or net battery current.

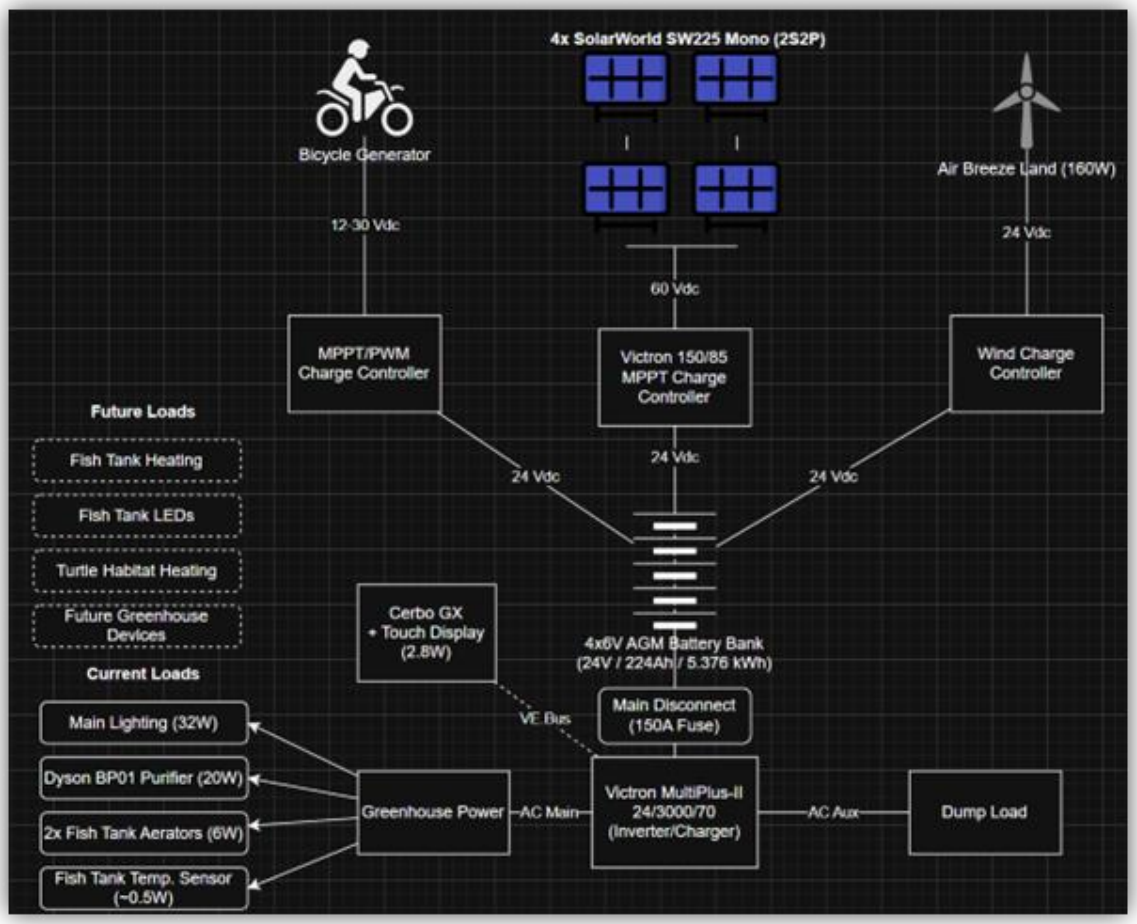


Fig 10. Energy flow chart

This system-level electrical diagram above provides an accurate description of the greenhouse as it stands today. Solar energy is harvested through four SolarWorld SW225 Mono panels arranged in a two series, two parallel configurations, delivering 60 Vdc to a Victron 150/85 MPPT charge controller. Supplementing solar generation is a 160W Air Breeze Land wind turbine, managed by a dedicated wind charge controller, and this project's bicycle generator producing 12–30 Vdc, routed through an additional MPPT/PWM charge controller. All three sources converge on a 4x6V AGM Battery Bank (24V / 224Ah / 5.376 kWh), which serves as the system's primary energy storage.

From the battery bank, power flows through a 150A Main Disconnect fuse to a Victron MultiPlus-II 24/3000/70 Inverter/Charger, which supplies AC power to the greenhouse loads, and routes excess energy to a dump load via an AC auxiliary output. Current loads include main lighting (34W), a Dyson BP01 air purifier (20W), two fish tank aerators (6W), and a fish tank temperature sensor (~0.5W). System monitoring and control is handled by a Cerbo GX with Touch Display (2.8W), connected via VE.Bus. Several future loads will be added, including fish tank heating and LEDs, turtle habitat heating, and other additional greenhouse devices, reflecting the system's design for scalability as energy needs increase.

d. Electrical Installation

As this capstone shifts from analysis and planning to installation and building, we are tasked with the installation of four new electrical receptacles and one new light switch in the greenhouse. Electrical energy generated by the solar panels and wind turbine is regulated and stored in the battery bank before being supplied to the inverter. The inverter converts the battery system's DC electrical energy to AC power for the greenhouse distribution system. The proposed receptacles and switch circuit extend this AC distribution to locations where additional electrical load may be required.

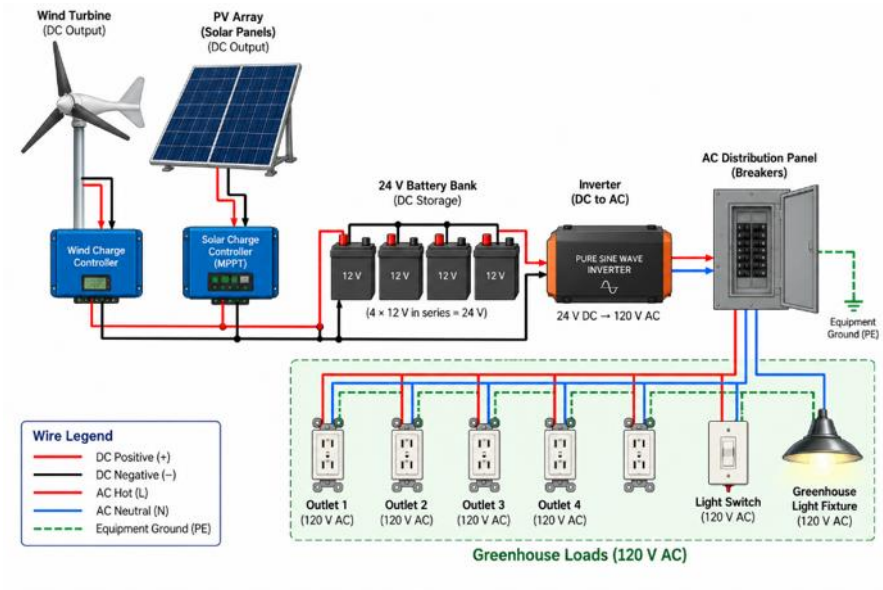


Fig 11. Conceptual Electrical System Configuration

The figure above serves as the electrical modeling component of this assignment by illustrating the complete path between renewable energy generation and the proposed loads. The model also demonstrates that adding receptacles and the light switch affects more than the final AC wiring. The battery bank, inverter capacity, circuit protection, conductors, and anticipated electrical loads must also be considered when evaluating the feasibility of the installation.

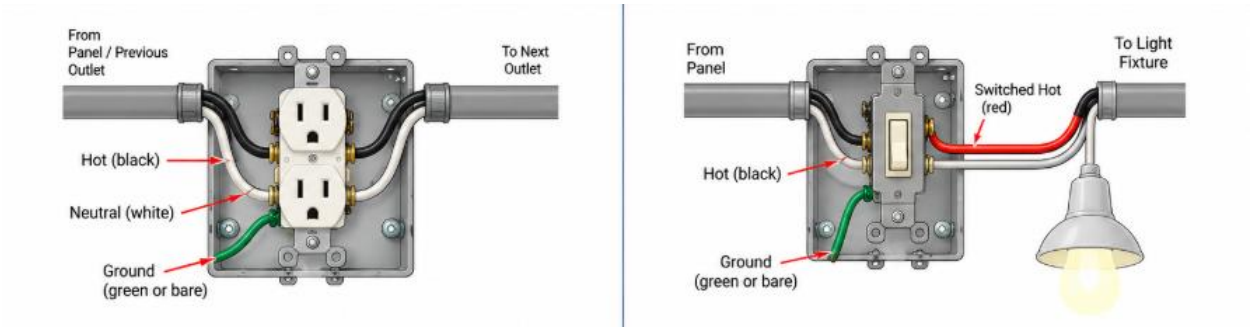


Fig 12. Conceptual Wiring Representation

To better understand and support the installation of the electrical receptacles and switch the above wiring representation will be used.

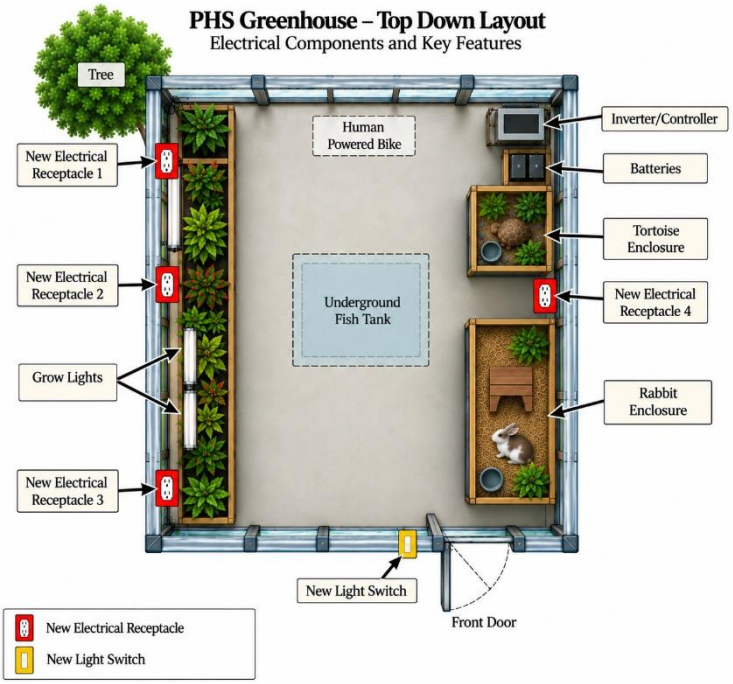


Fig 13. Top-down diagram of the electrical receptacle and light switch installation locations

Four receptacle outlets are proposed at locations within the greenhouse as shown in the figure above. These locations were determined based on physical greenhouse layout, anticipated equipment use, accessibility, environmental exposure, and applicable electrical requirements.

VI. Moving Forward

After completion of the spring semester and beginning of the fall semester for the PHS sustainability capstone, the team has completed all necessary calculations, diagrams, and designs apart from the additional solar panel design. As previously discussed with Prof. Pete, the solar panel design will be generated in the 66% hardware update with help from the renewable energies course (ME451). The team has confirmed designs with the clients and Prof. Pete. Additionally, all designs have been validated through hand calculations, SolidWorks analysis, and MATLAB code.

VII. Conclusion

At this time, all necessary calculations for the project have been completed. The team can now move forward with implementing their designs. This document overviews the mathematical verifications for the designs and will be used for reference throughout the manufacturing and installation process.

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Appendices

Appendix A: Tables and Figures for the Greenhouse Solutions

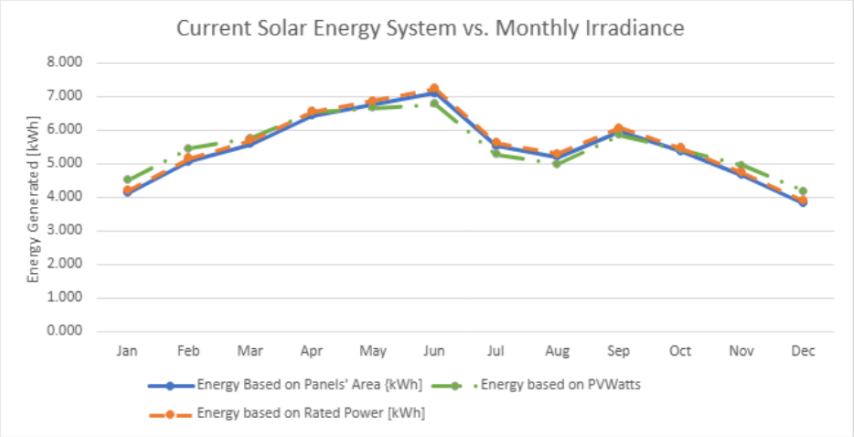


Fig A1. Current Solar Energy System vs. Monthly Irradiance

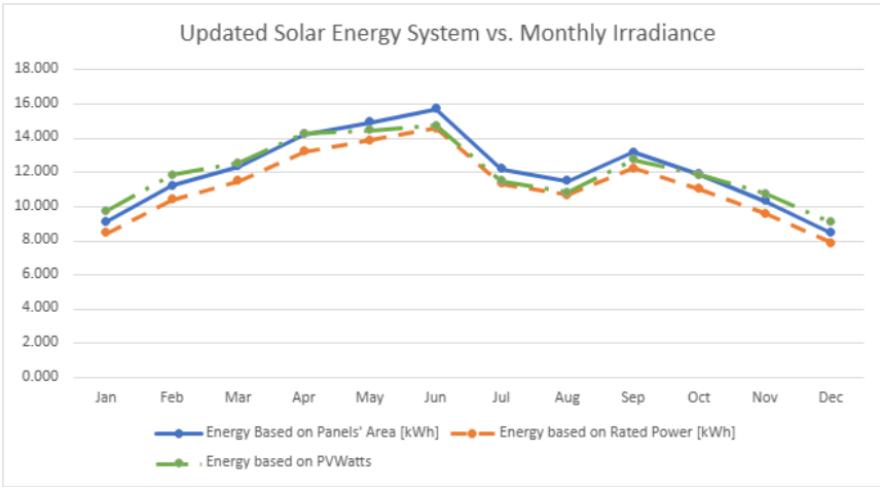


Fig A2. Updated Solar Energy System vs. Monthly Irradiance

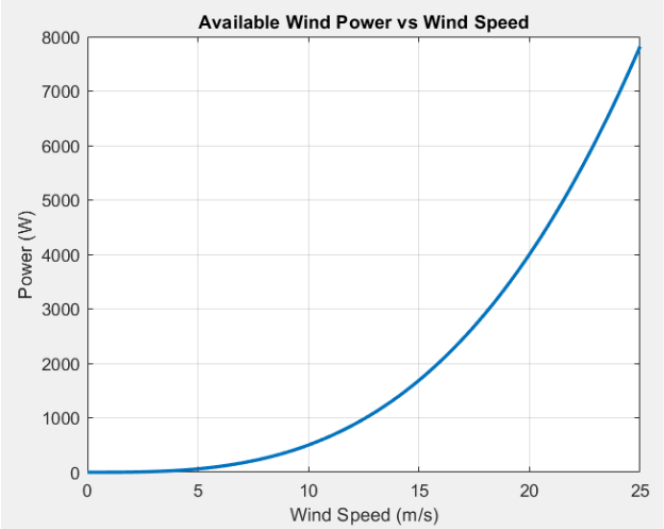


Fig A3. Available Wind Power vs Wind Speed

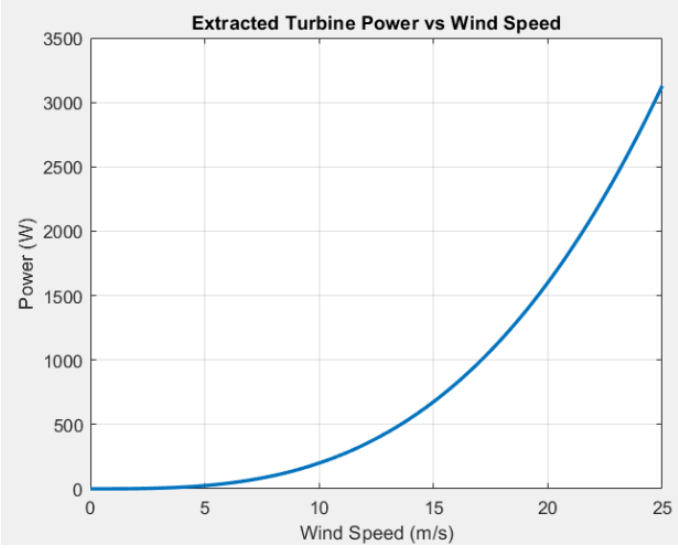


Fig A4. Extracted Turbine Power vs Wind Speed

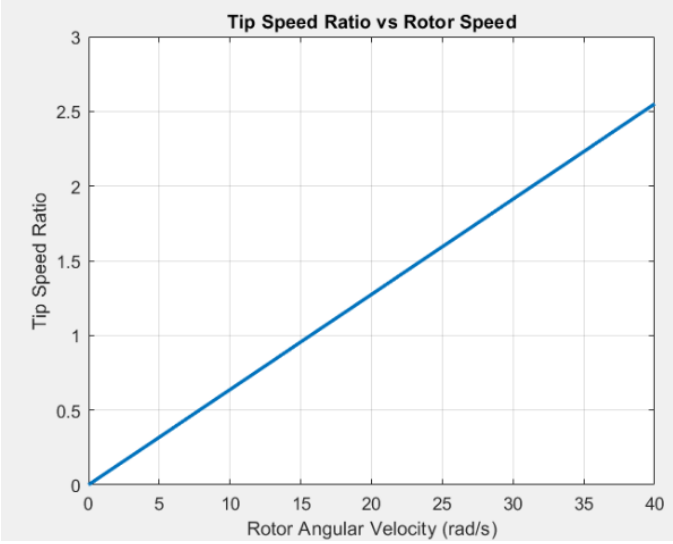


Fig A5. Tip Speed Ratio vs Rotor Speed

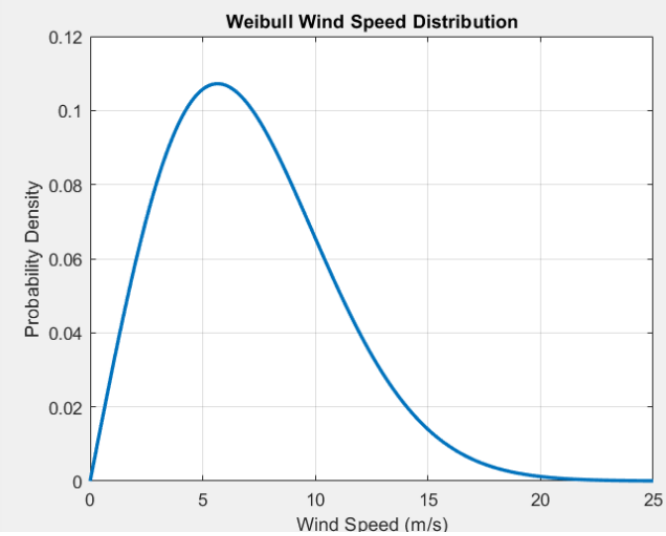


Fig A6. Weibull Wind Speed Distribution

Table AI

Total Power Consumption of Tortoise Enclosure

Power Total			
W	High	Avg	Low
Winter	50.28762	172.3048	305.3791
Spring/Fall	1.209179	119.088	231.1084

Summer	0	49.31769	178.4477

Table AII

Energy Consumption, No Lamp or Pond

Energy (kWh) No Lamp or Evaporation			
W	High	Avg	Low
Winter	72.10453	247.0579	437.8656
Spring/Fall	1.733772	170.7534	331.373
Summer	0	70.7138	255.8659
Total	73.8383	488.5251	1025.104

Table AIII

Total Energy of the Tortoise Enclosure

Energy (kWh) Lamp and Water			
W	High	Avg	Low
Winter	299.997	474.9504	665.7581
Spring/Fall	229.6263	398.6459	559.2655
Summer	227.8925	298.6063	483.7584
Total	757.5158	1172.203	1708.782

Table AIV

Battery Calculations for Comparison

Battery	**Company /Place for Purchase**	**Ah Capacity (Ah)**	**kWh Capacity (kWh)**	**Parallel or Series**	**System Voltage (V)**	**Cont. Discharge Current (A)**
DC224-6 (Current)	[2]	224	1.344	Series and/or Parallel	6	11.2

Lead Acid based (AGM)						
FHSKY-24051-G2-INV LiFePO4	[3]	200	5.12	Parallel (up to 16 times)	24	200
GWN12200 Sodium ion	[4]	200	2.4	Series and/or Parallel (4P4S)	12	200
GWN12100 Sodium ion	[4]	100	1.2	Series and/or Parallel (4P4S)	12	100
BAT524120740 LiFePO4	[5]	100	2.56	Parallel	25.6	200

Table AV

Heat Generation from Human Powered Bike

Human Watts	75	150	350	400
Power (W)	45.75	91.5	213.5	244
current (A)	3.13	6.27	14.62	16.71
Heat Gen (W)	0.39	1.57	8.55	11.17

Table AVI

Estimated Temperatures Based on Season

Celsius	Winter			
	Q1	Q2	Q3	Q4
T-high	19.05179	19.54132	22.43702	23.52394
T-Med	0.180294	0.669824	3.565517	4.65244
T-Low	-20.3982	-19.9087	-17.013	-15.9261

Spring/Fall				Summer			
Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
26.64179	27.13132	30.02702	31.11394	34.42079	34.91032	37.80602	38.89294
8.411794	8.901324	11.79702	12.88394	19.20179	19.69132	22.58702	23.67394
-8.91221	-8.42268	-5.52698	-4.44006	-0.76821	-0.27868	2.617017	3.70394

Appendix B: Tables and Figures for Human Powered Bicycle Solutions

Table BI

Energy Output Results

Gear	Teeth	ω (RPM)	F (N)	ω_alt (RPM)	P (W)
G1	11	45	250	1978	137.5
G2	13	48	239	1771	139.2
G3	15	50	228	1619	140.1
G4	18	54	212	1454	140.3
G5	22	60	190	1305	138.0
G6	26	65	168	1201	133.0
G7	34	75	125	1067	114.6

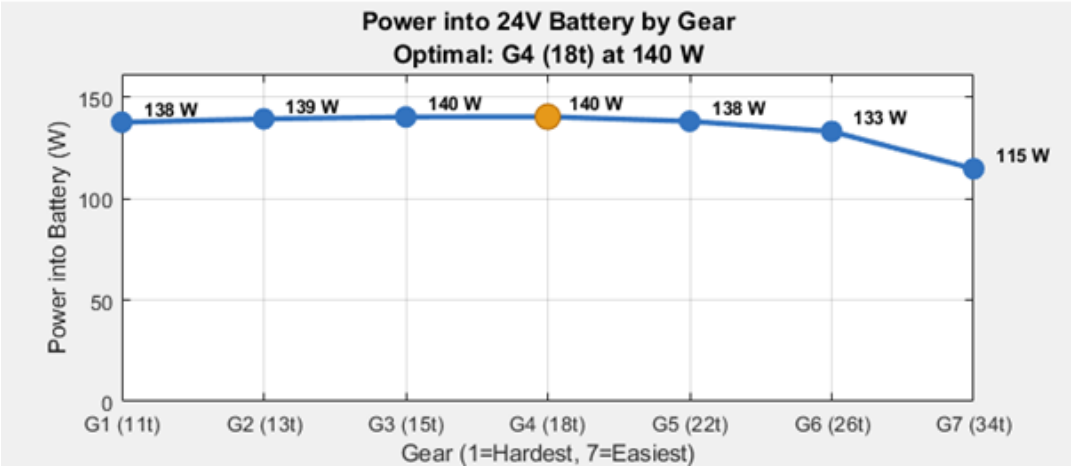


Fig B1. Power into 24V Battery by Gear

Table BII

Current Draw for HPE Bike Components

Component	Active Draw i_a (mA)	Standby Draw i_s (mA)	Wake Time t_w (ms)	Source
ESP32	11	0.8	12	[68]
7.5" E-Ink	7.9	0.001	200	[69]
INA219	1	0.006	1	[70]
Arduino Uno	50	-	-	[71]
16x2 LCD	25	-	-	[72]

Table BIII

RPM Results

T(s)	I_{avg} (mA)	P_{avg} (mW)	P_{avg} (Arduino & LCD)	P_{net}(W)	Power Improvement (%)
2	2.840	9.373	247.5	139.990	96.21
5	1.620	5.347	247.5	139.995	97.84
10	**1.213**	**4.005**	**247.5**	**139.996**	**98.38**
30	0.942	3.110	247.5	139.997	98.74
60	0.874	2.886	247.5	139.997	98.83

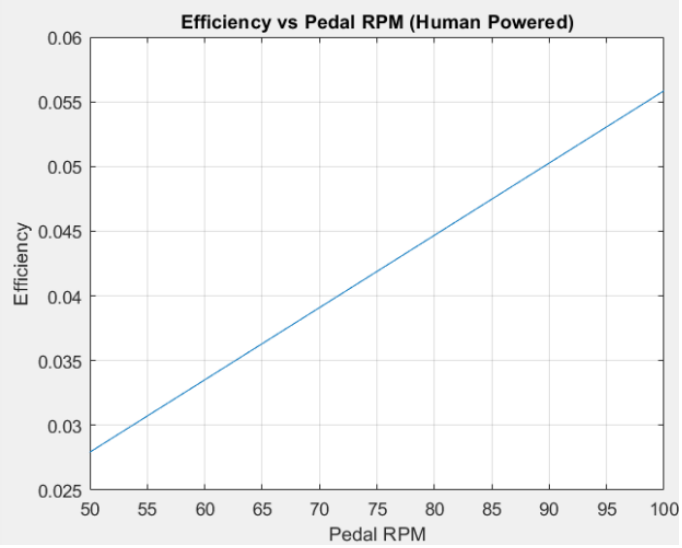


Fig B2. Efficiency vs Pedal RPM (Human Powered)

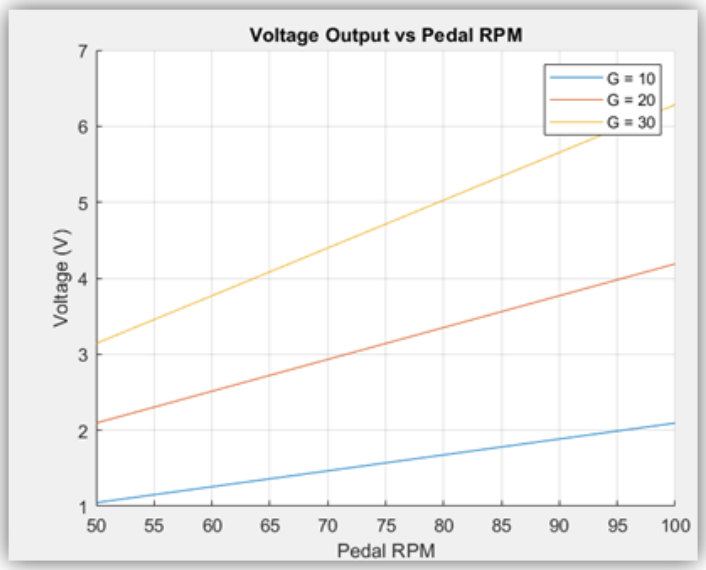


Fig B3. Voltage Output vs Pedal RPM

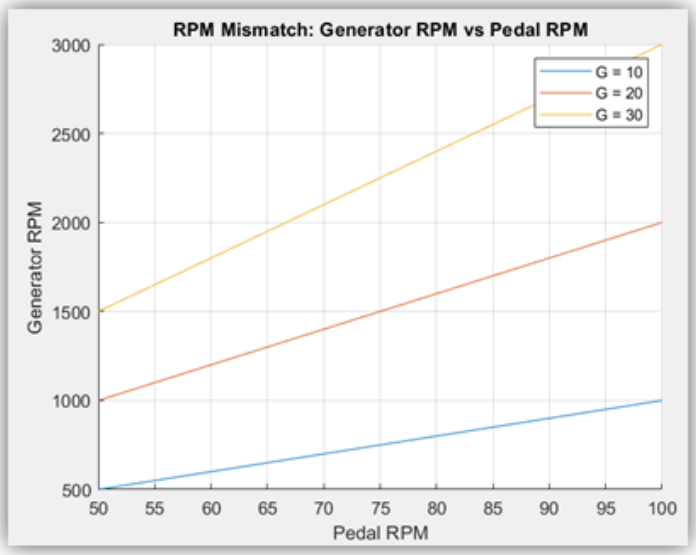


Fig B4. RPM Mismatch: Generator RPM vs Pedal RPM

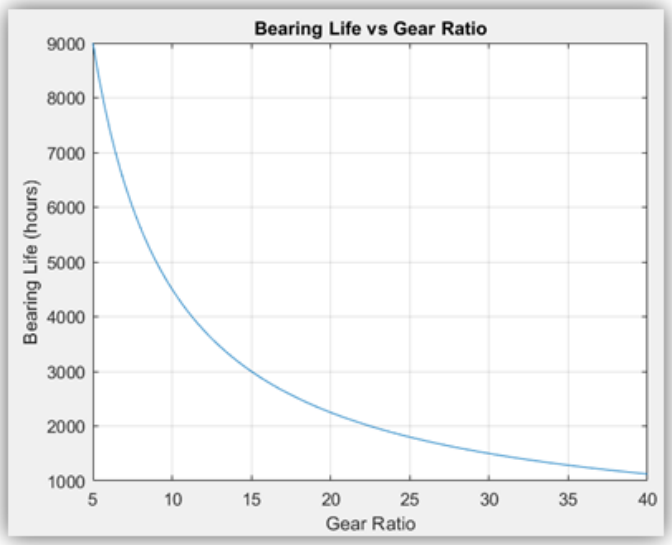


Fig B5. Bearing Life vs Gear Ratio

Appendix C: Factor of Safety Calculations

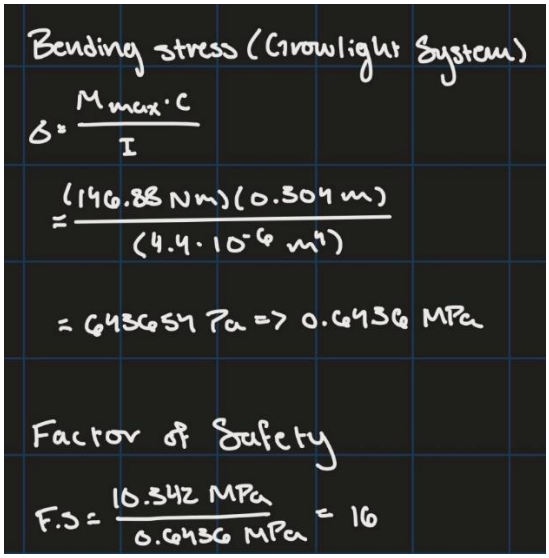


Fig C1. Grow-light system FoS calculations [6]

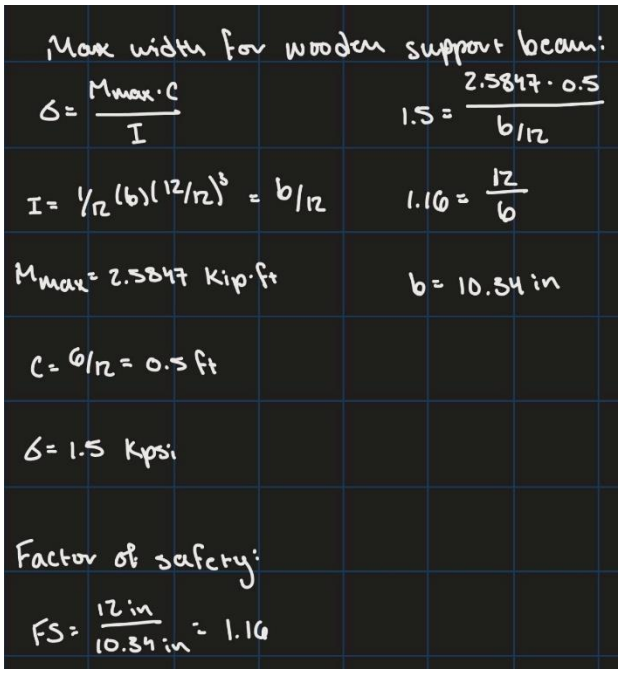


Fig C2. Solar panel loading FoS calculations [6]

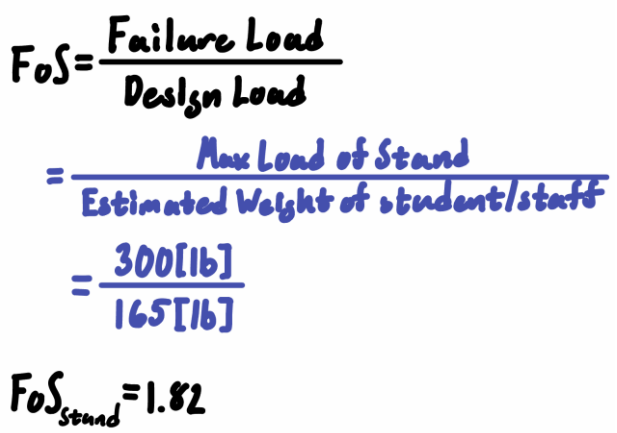


Fig C3. Bicycle Stand Load FoS calculations